```
from google.colab import files
import cv2
import numpy as np
import matplotlib.pyplot as plt

uploaded = files.upload()
filename = list(uploaded.keys())[0]
image = cv2.imread(filename, 0)
print("Image uploaded successfully:", filename)

kernel = np.ones((15, 15), np.uint8)

tophat = cv2.morphologyEx(
    image, cv2.MORPH_TOPHAT, kernel
)

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(image, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(tophat, cmap="gray")
plt.title("Top-Hat Output")
plt.axis("off")

plt.show()
```

Choose Files Screenshot… 075823.png

Screenshot 2026-10-10 075823.png(image/png) - 93416 bytes, last modified: 10/10/2026 - 100% done

```
Saving Screenshot 2026-10-10 075823.png to Screenshot 2026-10-10 075823 (1).png
Image uploaded successfully: Screenshot 2026-10-10 075823 (1).png
```



Welcome to Colab!

What is Colab?

Colab, or "Colaboratory", allows you to write and execute Python in your browser, with

- Zero configuration required
- Access to GPUs free of charge
- Easy sharing

Whether you're a **student**, a **data scientist** or an **AI researcher**, Colab can make your work easier. Watch Introduction to Colab or Colab Features You May Have Missed to learn more, or just get started below!